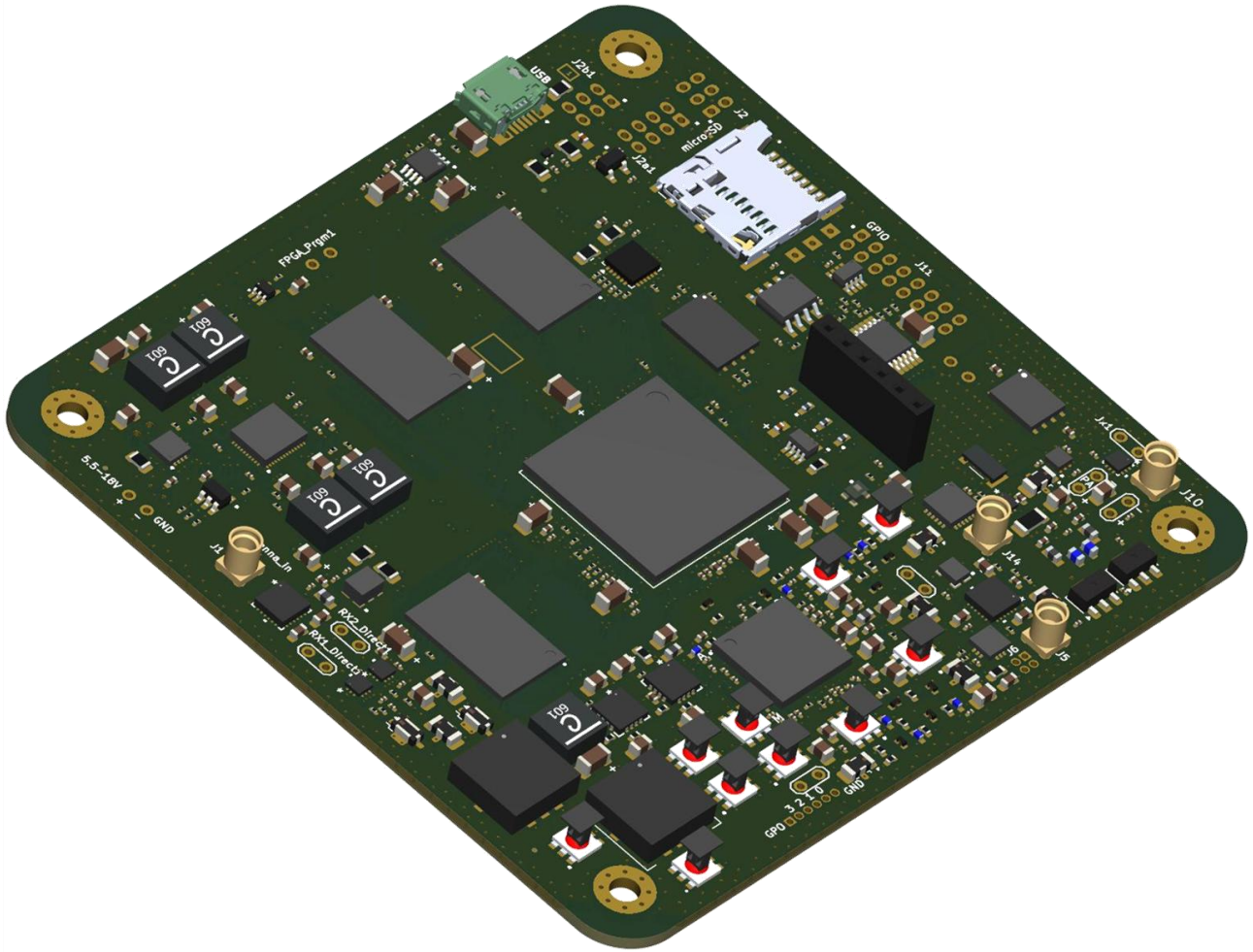
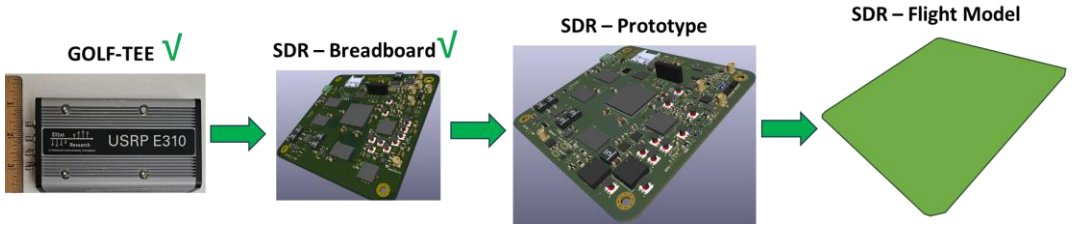
SDR GEN2 Prototype

A Versatile, Programmable (GNU Radio) Transceiver/Transponder – Designed for a 1U Footprint

- *144 MHz – 6 GHz Coverage, 10GHz exp.*
- *Direct VHF, UHF, 900 MHz, L, S, C and X Ops*
- *Up Conversion to "Five and Dime" Capability*
- *All Mode: CW, SSB, FT8, NBFM, SSTV, FSTV, etc.*
- *Downlink Rate up to 1Mbps*
- *Built to Fly on GOLF and Fox Plus Missions*
- *Open Source HW/SW*





AMSAT Engineers Keeping Amateur Radio in Space!

ASCENT – Advanced Satellite Communications & Exploration of New Technologies

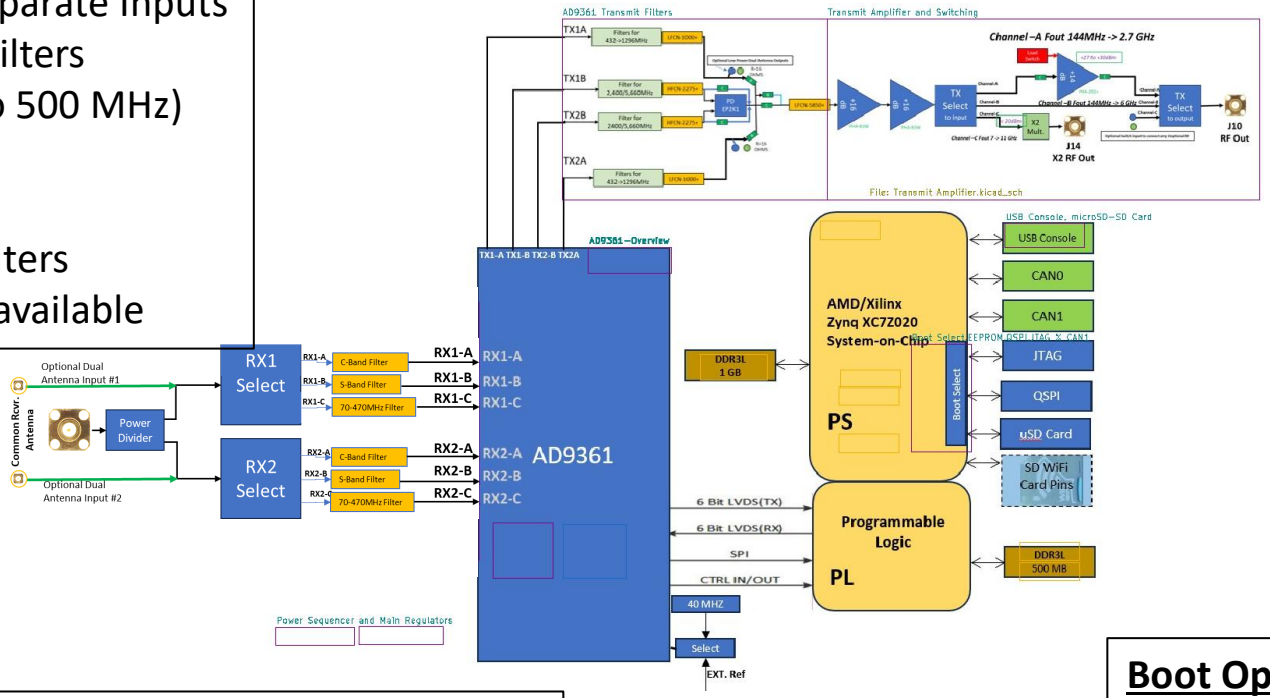
Follow AMSAT News Service
For More Information!

WA1CYB
KI5YNG

SDR GEN2 Prototype

2 Channel Receiver

- Common Input
  - Alternative Separate Inputs
- 3 Selectable Input Filters
  - VHF (2 mtrs to 500 MHz)
  - S Band
  - C Band
- Std Mini-Circuits Filters
  - Substitutions available



2 Channel Transmitter

- Common Output
  - Alternative Separate Outputs
- 2 Selectable Output Filters
  - VHF (2 mtrs to 1296 MHz)
  - S/C Band
- Std Mini-Circuits Filters
  - Substitutions available
- Output Power(s)
  - ½ to 1 watt (2 mtrs to 2700 MHz)
  - ~18 dBm (2 mtrs to 6000 MHz)
- Experimental X Band Output
  - ~ 5 dBm power output

Xilinx Zynq 7020 FPGA

- 1GB System RAM
  - Linux
- ½ GB Programmable Logic Available for interfaces
  - UHD/ IIO Drivers planned
- Dual CAN Bus Interface
- Serial/USB Interface

Boot Options:

- JTAG
- Micro SD Card (64GB)
- QSPI Flash (128Mbit, 16 MByte) SPI Serial
- EEPROM (256K (32 kByte) I2C CMOS Serial)

Other: Optional Clk. Ref, Load Switch for PA, GPIO

WA1CYB
KI5YNG

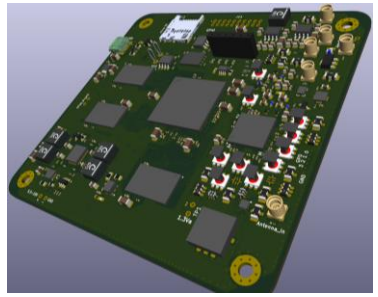
Backup

Hamvention 2026

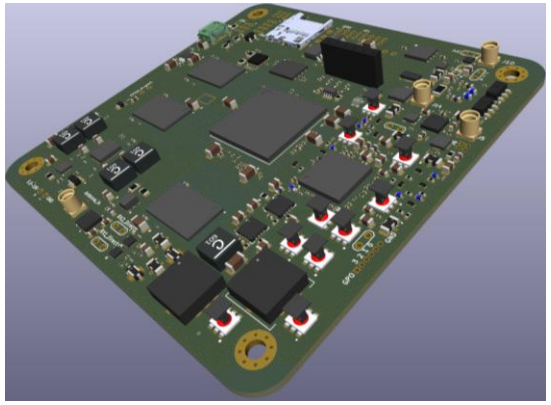
SDR Transponder/Transceiver for AMSAT

GOLF-TEE √

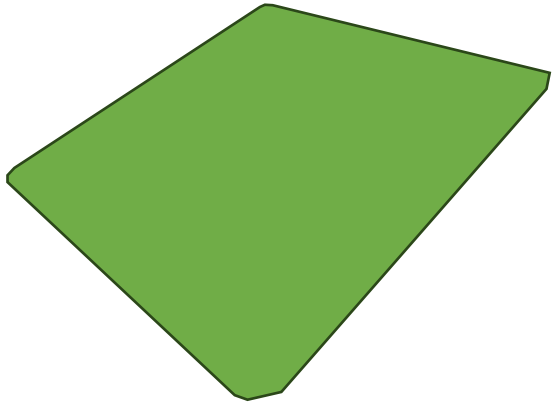
SDR – Breadboard √



SDR – Prototype



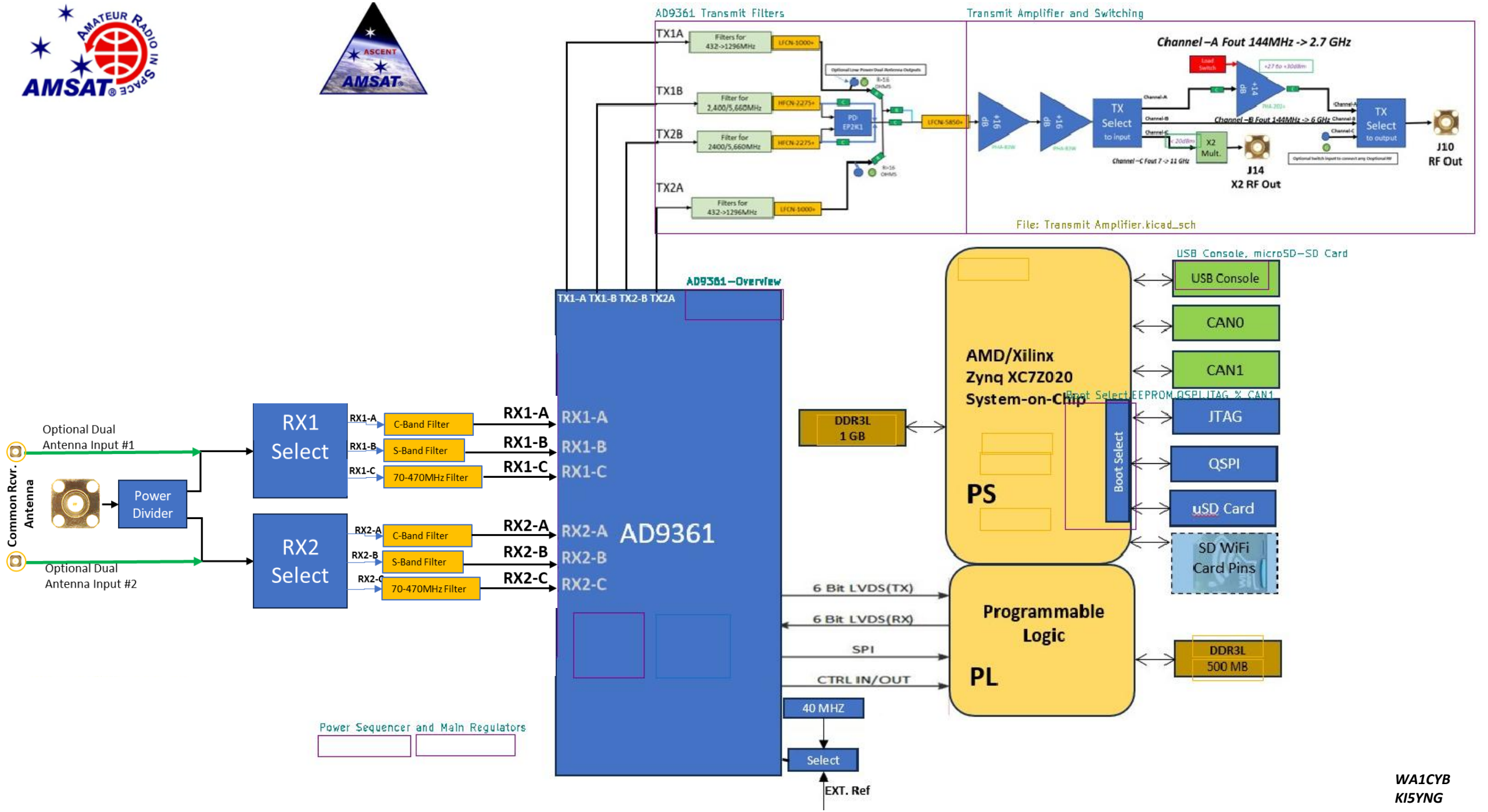
SDR – Flight Model



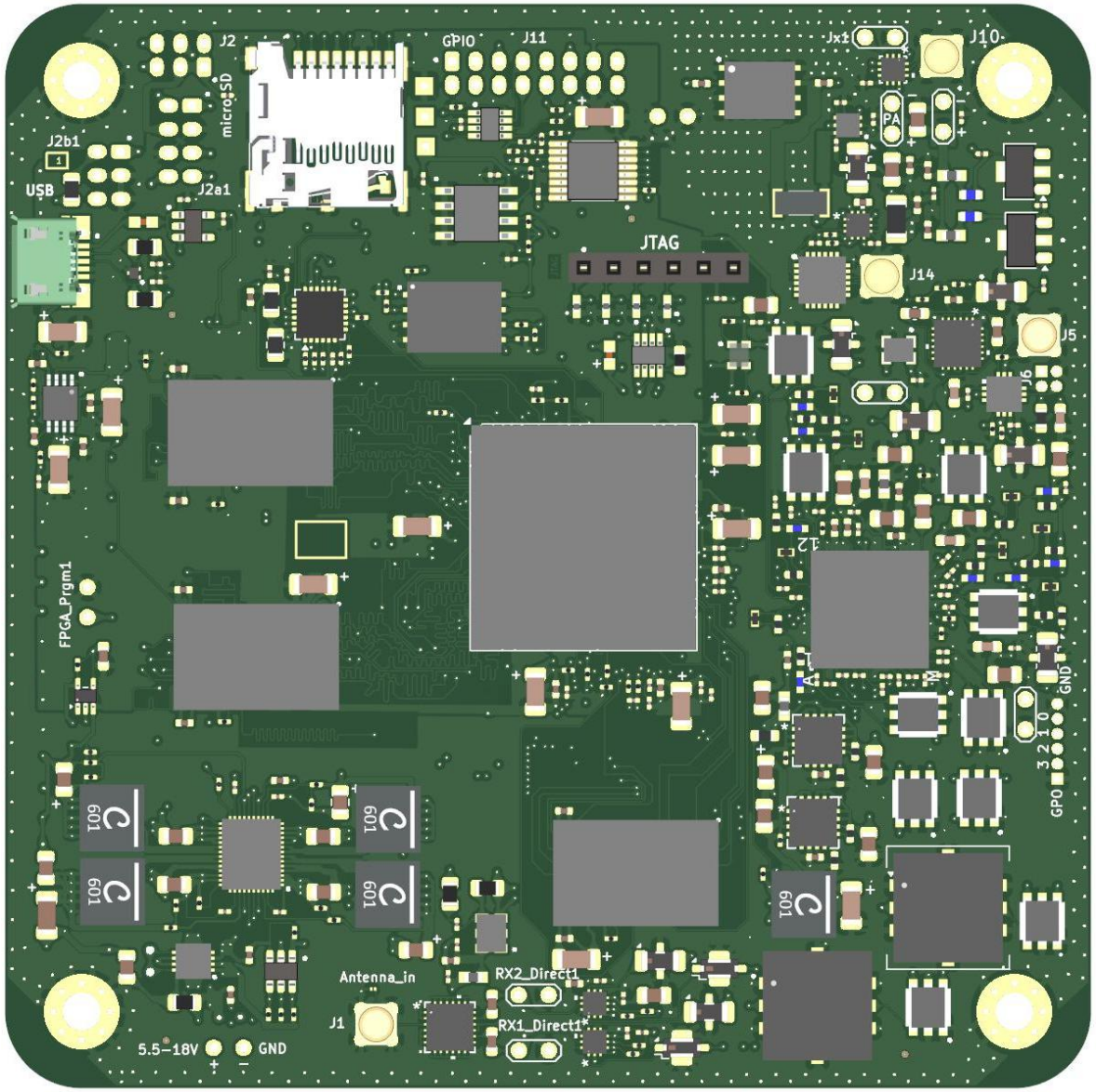
AMSAT Engineers Keeping Amateur Radio in Space!

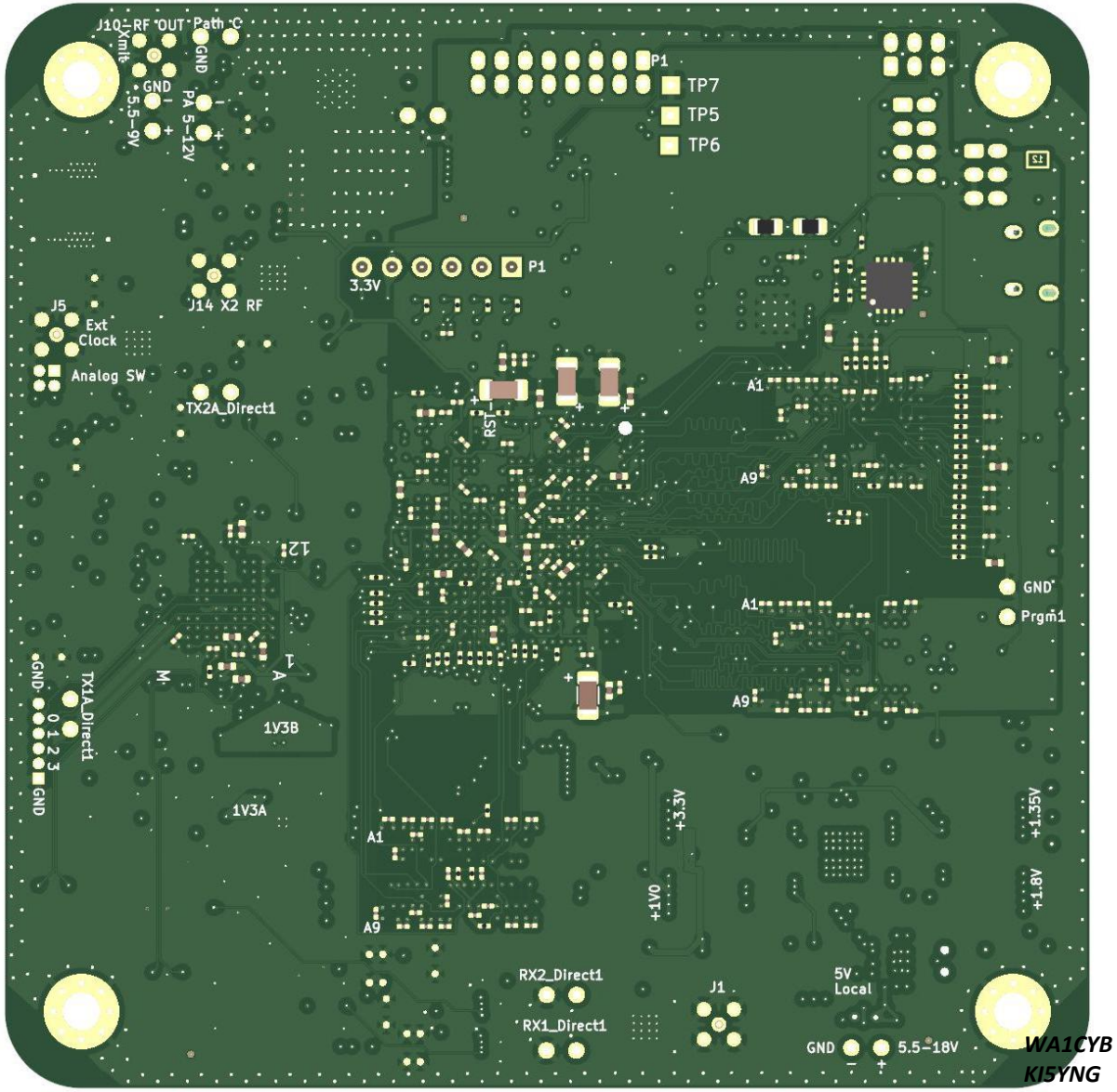
ASCENT – Advanced Satellite Communications & Exploration of New Technologies

WA1CYB
KI5YNG


AD9361 Transmit Filters
Transmit Amplifier and Switching
TX1A
TX1B
TX2B
TX2A
Filters for 432->1296MHz
Filter for 2,400/5,660MHz
Filter for 2400/5,660MHz
Channel –A Fout 144MHz -> 2.7 GHz
TX Select to input
TX Select to output
X2 Mult.
J14
X2 RF Out
J10
RF Out
File: Transmit Amplifier.kicad_sch
AD9361–Overview
TX1-A TX1-B TX2-B TX2A
RX1-A
RX1-B
RX1-C
RX2-A
RX2-B
RX2-C
AD9361
USB Console, microSD–SD Card
USB Console
CAN0
CAN1
JTAG
QSPI
uSD Card
SD WiFi Card Pins
AMD/Xilinx Zynq XC7Z020 System-on-Chip
PS
Boot Select
DDR3L 1 GB
Programmable Logic
PL
DDR3L 500 MB
6 Bit LVDS(TX)
6 Bit LVDS(RX)
SPI
CTRL IN/OUT
40 MHZ
Select
EXT. Ref
Common Rcvr. Antenna
Optional Dual Antenna Input #1
Optional Dual Antenna Input #2
Power Divider
RX1 Select
RX2 Select
C-Band Filter
S-Band Filter
70-470MHz Filter
Power Sequencer and Main Regulators
WA1CYB
KI5YNG

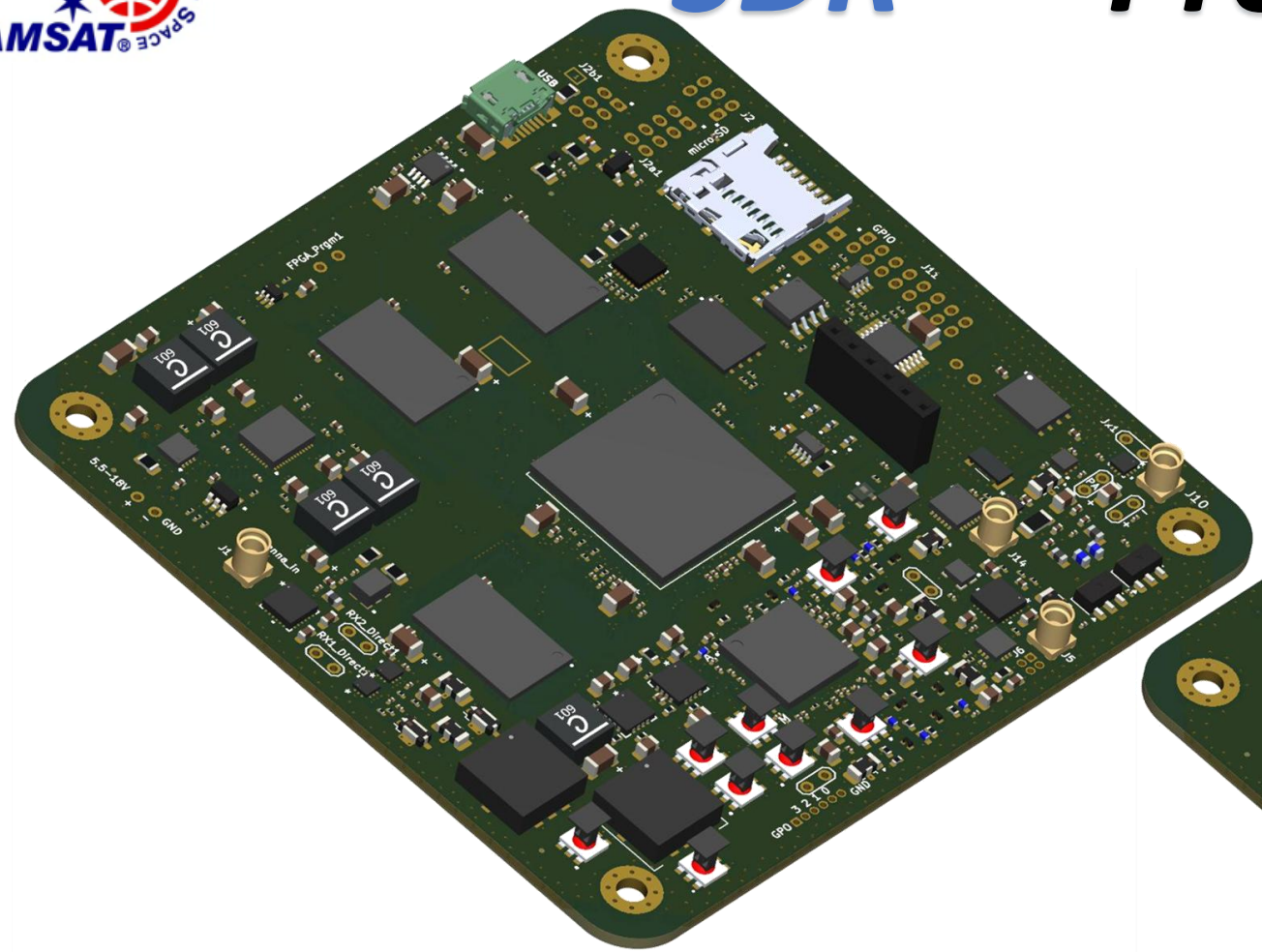
SDR GEN2 Prototype

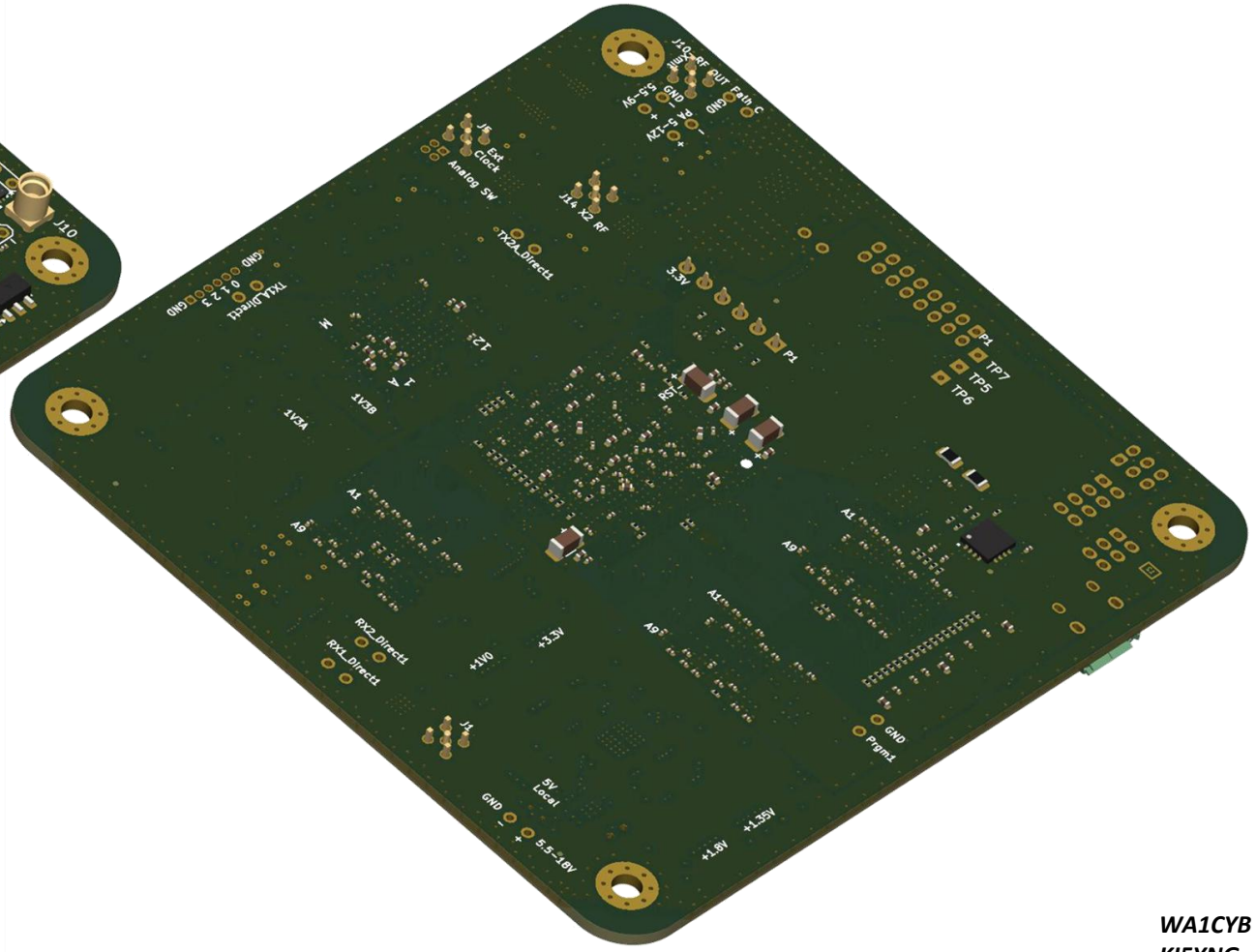




WA1CYB
KI5YNG

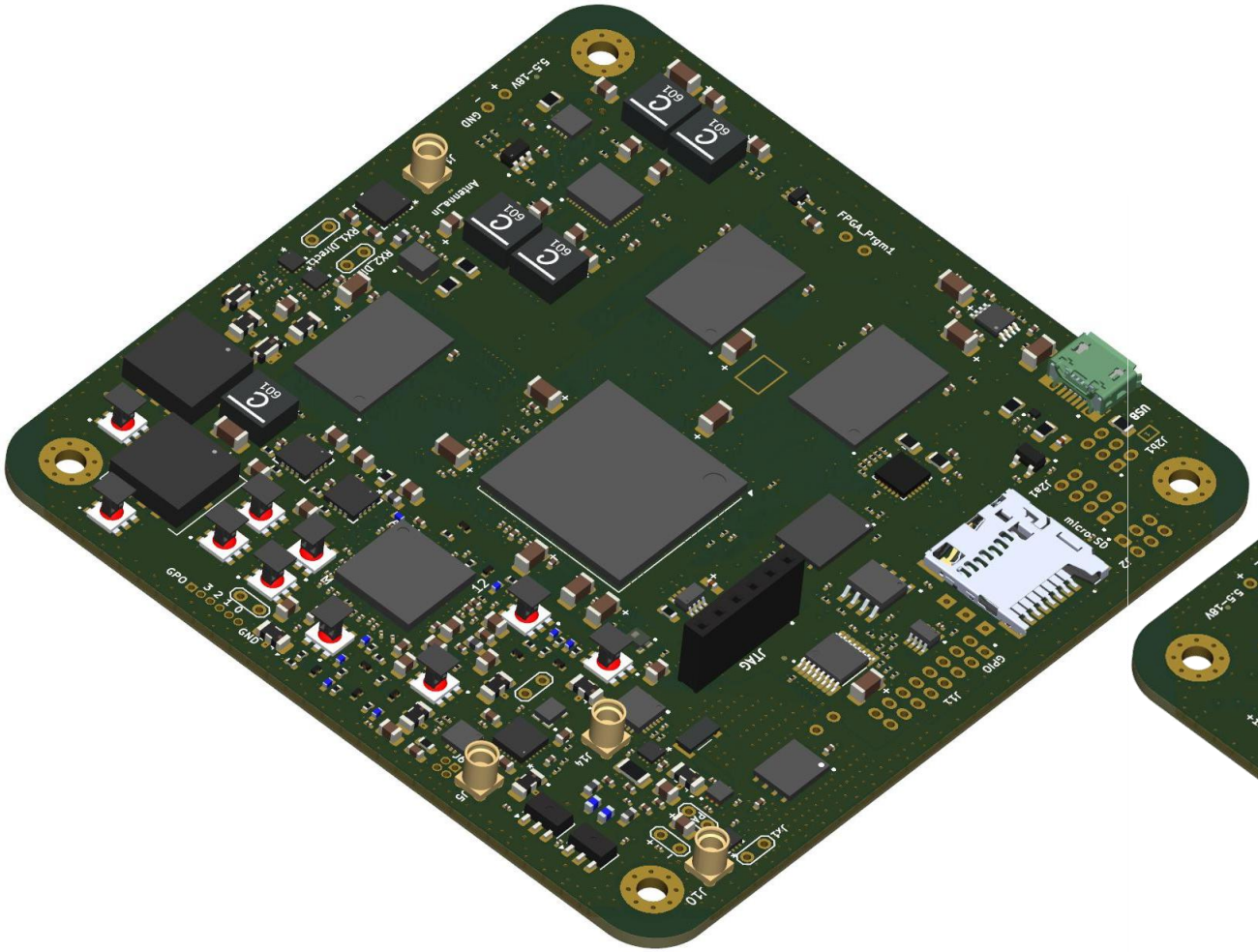
SDR GEN2 Prototype

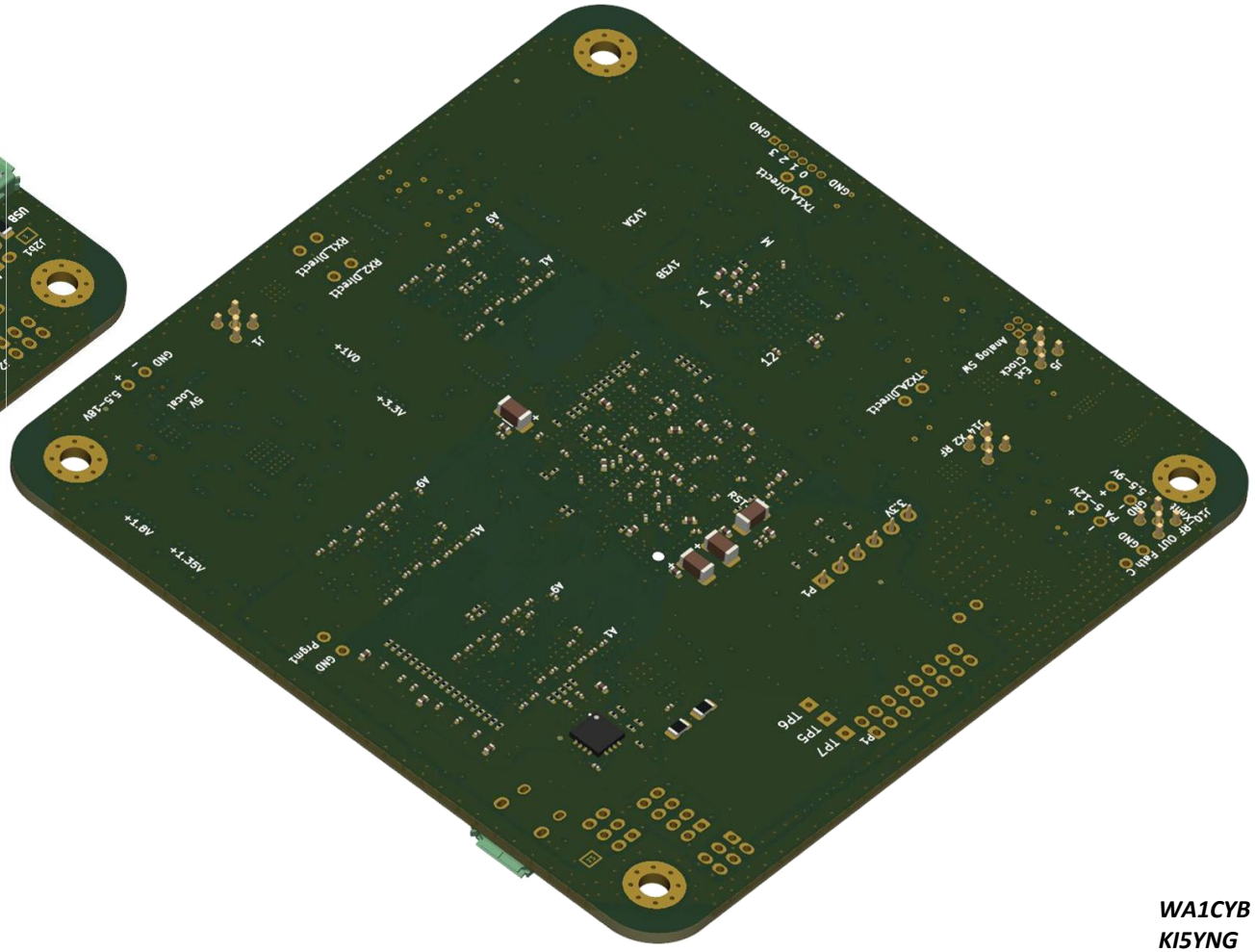




WA1CYB
KI5YNG

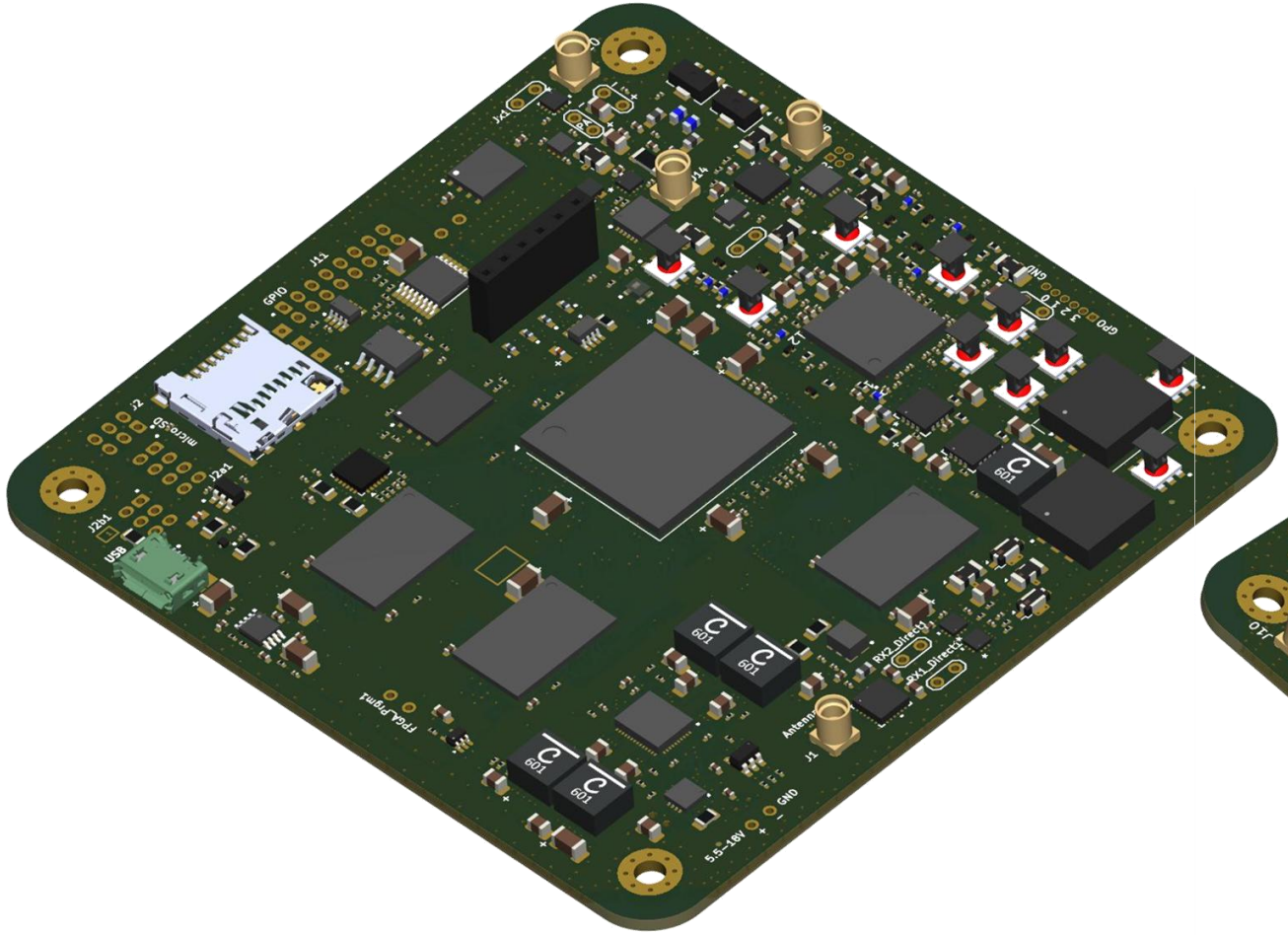
SDR GEN2 Prototype

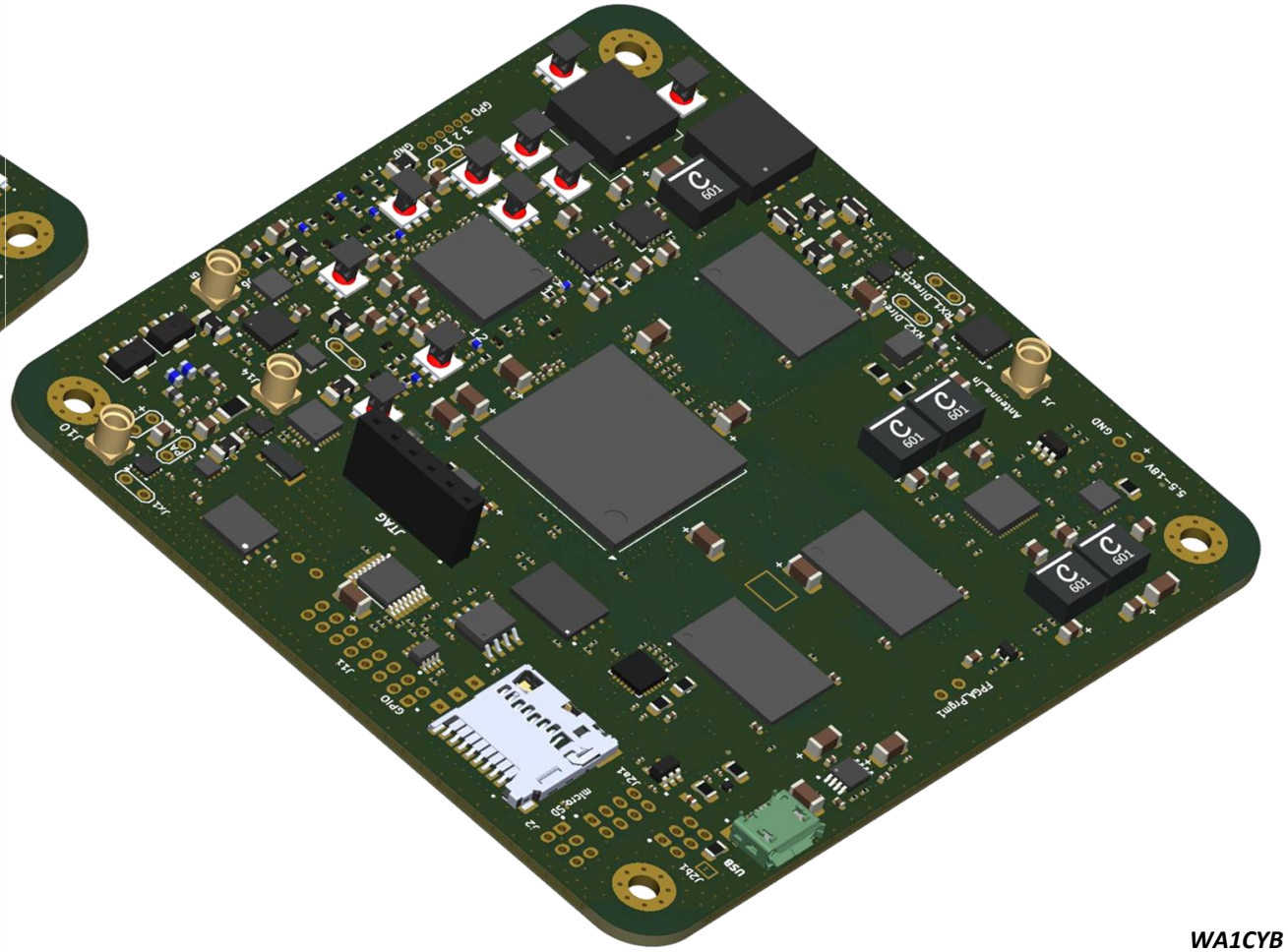




WA1CYB
KI5YNG

SDR GEN2 Prototype





WA1CYB
KI5YNG

- Replaces the Ettus E310 (**123.9 x 68.2 x 26.4 mm**) with a custom SDR card (**92 x 92 x7.5 mm + Heat Sink**)
- Provides 1 watt output (goal)
- Open source (It will be / is on GitHub) for now it will be in: **https://github.com/WA1CYB/SDR_Gen2**
- KiCad 9 schematic and layout used. Re-use and modifications encouraged